

Number, units and estimation - Set 2

TMUA practice questions by topic. Specification M1.1, M1.2, M2.1-M2.14. 9 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume 3.6 m^3 ? ($1 \text{ m}^3 = 1000 \text{ litres}$.)
- A 30 minutes
B 50 minutes
C 5 minutes
D 1 hour
E 3 hours
F 500 seconds
- 2 $a = 5.0$ and $b = 2.0$, each correct to 2 significant figures. Find the lower bound of $\frac{a}{b}$.
- A $\frac{4.5}{2.5}$
B $\frac{4.5}{2.05}$
C $\frac{4.05}{2.05}$
D $\frac{4.95}{2.05}$
E $\frac{2.4}{5.05}$
F $\frac{2.05}{2.05}$
- 3 What is the smallest positive integer that is divisible by every integer from 1 to 10?
- A 3628800
B 1260
C 840
D 720
E 5040
F 2520

- 4 Which of the following fractions has a terminating decimal expansion?
- A $\frac{7}{40}$
 - B $\frac{9}{22}$
 - C $\frac{11}{14}$
 - D $\frac{14}{9}$
 - E $\frac{5}{12}$
- 5 Find the exact value of $(\sqrt{5} - 1)^2 + (\sqrt{5} + 1)^2$.
- A 12
 - B 8
 - C 10
 - D 6
 - E $12 + 4\sqrt{5}$
 - F $4\sqrt{5}$
- 6 The lowest common multiple of 12, 18 and n is 180. Which of the following could be n ?
- A 20
 - B 24
 - C 8
 - D 40
 - E 50
 - F 27
- 7 How many positive divisors does 360 have?
- A 24
 - B 20
 - C 18
 - D 36
 - E 12
 - F 16

- 8** n is a positive integer such that n , $n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I** There is exactly one such value of n .
- II** n must be odd.
- III** One of n , $n + 2$, $n + 4$ must be divisible by 3.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 9** n is a positive integer greater than 1. Which of the following statements is/are true?

- I** If n is not prime, then n has a prime factor at most $\sqrt{n}$.
- II** If n has no prime factor at most $\sqrt{n}$, then n is prime.
- III** If n is prime, then n has no factor between 1 and n exclusive.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III